EXPERIMENT 3

Survey Report on Emerging Trends in Low-Code Application Development

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Project: Patient Appointment Booking System

Aim

To study and analyse the emerging trends in low-code application development and understand their impact on modern software development.

Theory

Low-code application development enables users to create applications using visual interfaces, drag-and-drop components, and minimal hand-written code. It accelerates application development, reduces costs, and enables professional developers and citizen developers to build business applications efficiently. Major emerging trends include:

Emerging Trend	Description
Artificial Intelligence (AI) Integration	AI-powered code suggestions, workflow automation, chatbots, and predictive analytics can improve development efficiency.
Citizen Development	Non-technical users can build business applications with minimal programming knowledge.
Cloud-Native Development	Cloud-based platforms support deployment, scalability, and remote collaboration.
Business Process Automation	Workflow automation streamlines repetitive business tasks.
Cross-Platform Application Development	Applications can be developed once and deployed across web and mobile platforms.
API and Third-Party Integration	Platforms can connect with external databases, cloud services, and enterprise applications.
Enhanced Security and Governance	Role-based access control, data encryption, audit logs, and compliance features help protect applications.
DevOps and Continuous Deployment	CI/CD integration enables faster testing, deployment, and application updates.

Procedure

1. Study the concept of low-code application development.

2. Identify popular low-code platforms and their key features.

3. Collect information on emerging trends from reliable sources such as research papers, industry reports, and official documentation.

4. Analyze trends such as AI integration, citizen development, cloud computing, automation, API integration, and security.

5. Compare the benefits and challenges associated with each trend.

6. Record the observations and summarize the findings.

7. Prepare the survey report based on the collected information.

Benefits and Challenges

Benefits	Challenges
Faster application development and reduced coding effort.	Customization may be limited by platform capabilities.
Automation reduces repetitive manual work.	Complex workflows and integrations may require extra configuration or code.
Cloud access supports collaboration and deployment.	Security, governance, and platform costs must be managed carefully.
Visual tools let more users participate in development.	Large systems need careful performance and scalability planning.

Application to the Patient Appointment Booking System

Visual form building can support patient registration and appointment booking; workflow automation can support appointment status changes; and user permissions can help control access. Integrations may connect the application to external services when required. These are potential applications of the trends, not claims that every feature has already been configured.

Result

The survey revealed that low-code application development is evolving through AI-assisted development, cloud-native platforms, citizen development, business process automation, API integration, enhanced security, and continuous deployment. These advancements can improve development speed and reduce coding effort while requiring attention to customization, security, scalability, and cost.